Machine Learning Tutorial: Predicting Titanic Survival Using Logistic Regression and Decision Trees

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Github link: https://github.com/sai6265/individual-assignment

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Introduction

Machine learning is becoming more and more important in modern data analysis. It helps researchers and professionals find patterns, make predictions, and help people make decisions in many different fields. The Titanic dataset is one of the most popular starting points for machine learning. It has information about the passengers on the RMS Titanic and whether they lived through the disaster. This dataset is very useful because it has demographic, social, and economic variables all in one place. This makes it possible to do meaningful analysis of real-world outcomes.

The goal of this report is to show how machine learning can be used to figure out how likely a passenger is to survive. This study specifically examines Logistic Regression and Decision Tree models, which are two commonly employed classification methods. The report stresses exploratory data analysis, feature importance, and model evaluation in addition to making predictive models. The analysis sheds light on the most important factors that affected survival during the Titanic disaster by looking at both statistical patterns and model outputs.

Dataset Description

The Titanic dataset has a lot of different variables that tell you about the passengers and their situations. The main target variable is survival status, which is binary and shows whether a passenger lived or died. There are also a number of explanatory variables in the dataset, such as age, sex, passenger class, fare, number of siblings or spouses on board, and number of parents or children on board.

These features give a mix of demographic and socioeconomic data. For instance, the class of a passenger and the fare they pay can show how rich or poor they are, while their age and family ties can show something about their personality. Because the dataset lets models learn how these variables are related to survival outcomes, it is a good fit for classification tasks.

Exploratory Data Analysis

Exploratory data analysis is a necessary part of machine learning because it helps find patterns, spot outliers, and learn about the data's structure before modelling (James et al.,

2021). This analysis employed various visualisations to examine distributions and correlations among variables.

The age distribution shows that most of the passengers were between 20 and 40 years old, and there weren't many older people. The distribution is slightly skewed to the right, which means that the dataset is mostly made up of younger passengers, but there are also a lot of older passengers. This means that age may not be evenly spread out in the population and could have a small but noticeable effect on survival.

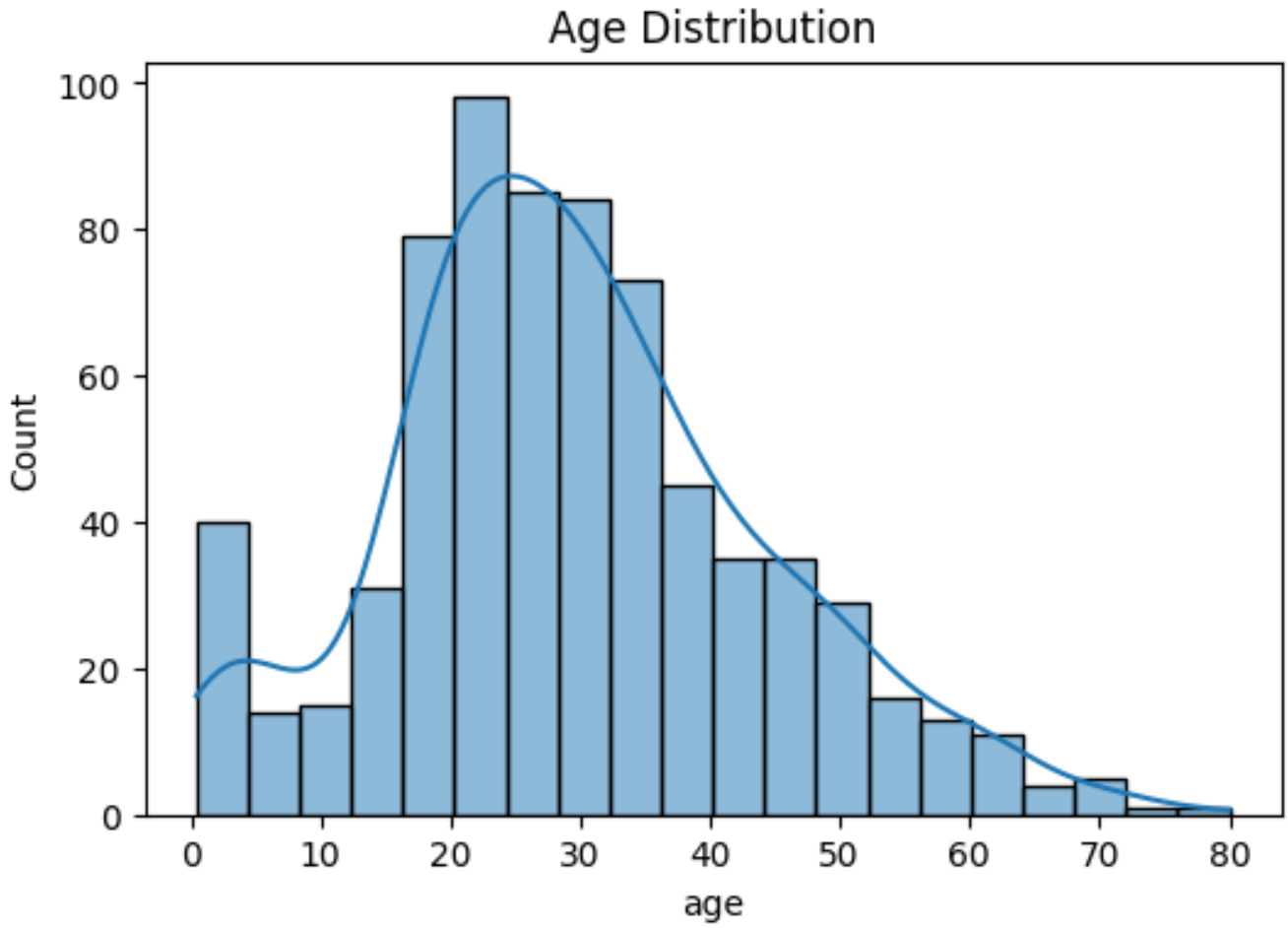

Figure 1: Distribution of passenger ages showing a concentration of individuals between 20 and 40 years.

The fare distribution is very uneven, with most passengers paying low ticket prices and only a few paying much higher prices. This pattern shows how the Titanic's class system worked, with first-class passengers paying a lot more than those in lower classes. The presence of extreme values suggests that wealth disparities were considerable and may have affected access to safety resources.

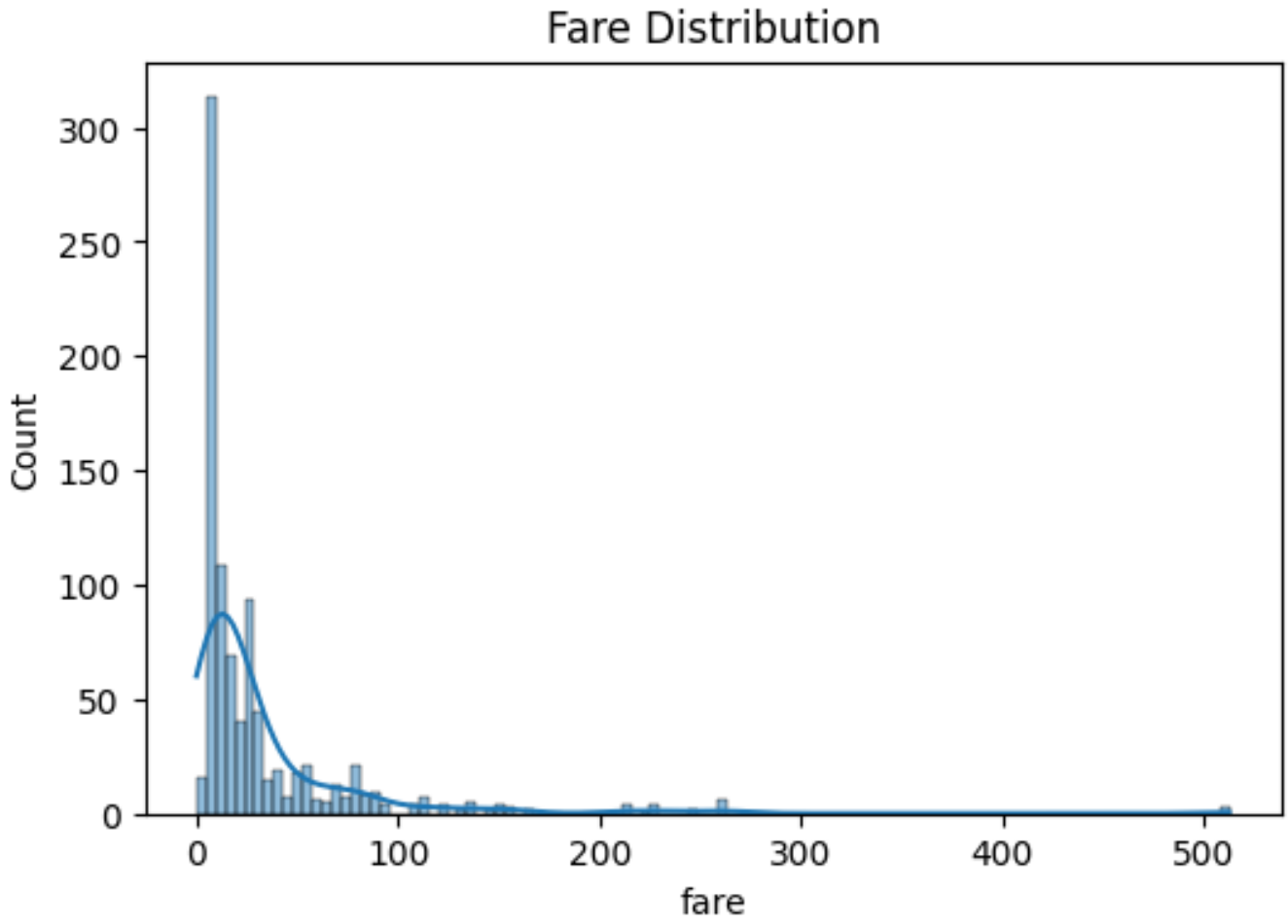

Figure 2: Distribution of passenger fares showing a highly right-skewed pattern with a few extreme values.

When looking at survival by sex, a clear trend can be seen. Women who were on the plane had a much better chance of living than men who were on the plane. This difference is big enough to show that gender was a major factor in determining who lived and who died. Historical accounts of the Titanic disaster show that women and children were given priority during evacuation, which explains why this happened.

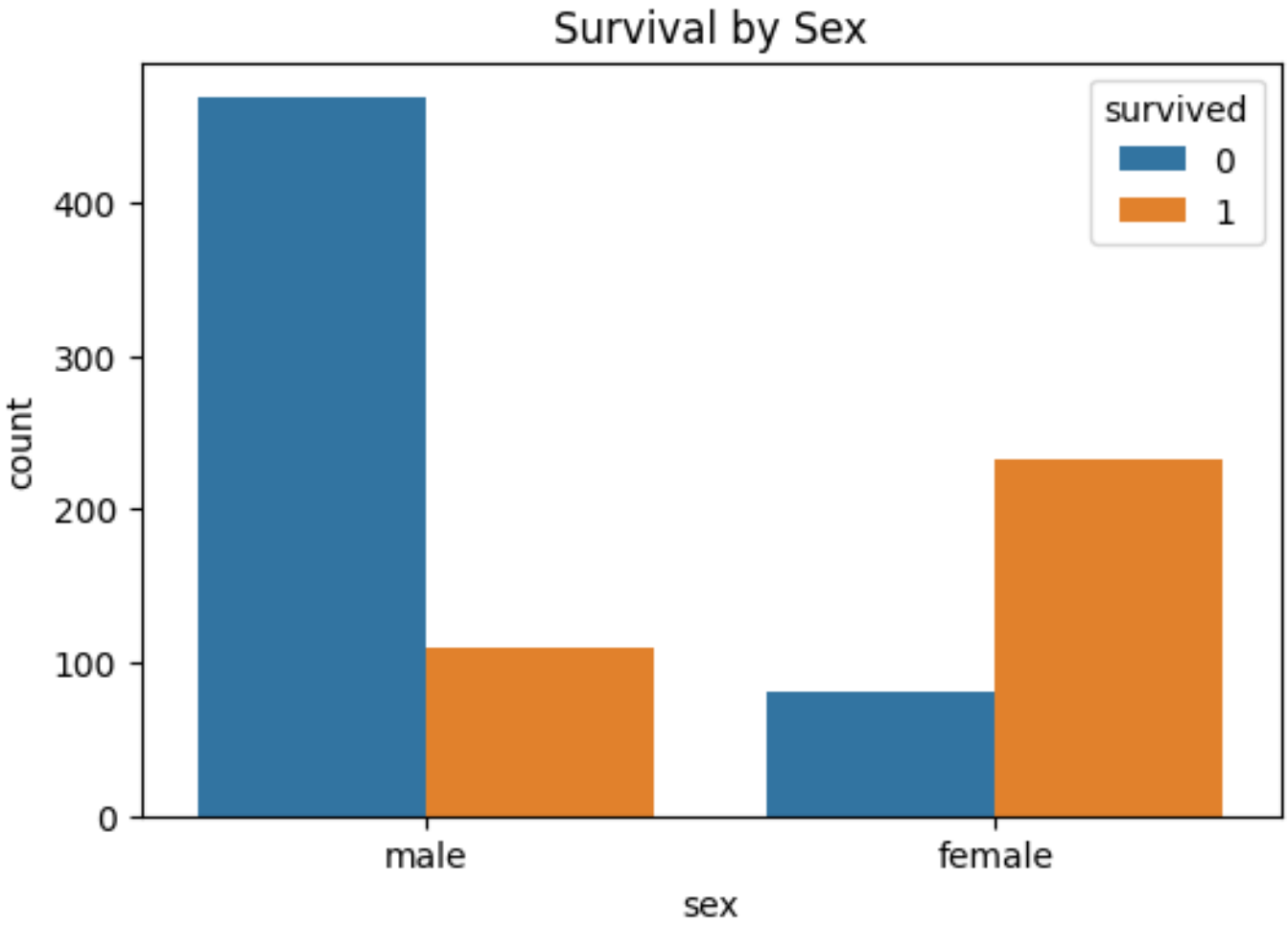

Figure 3: Survival counts by gender, highlighting significantly higher survival among female passengers.

There is also a strong link between passenger class and survival. The first-class passengers had the best chances of survival, followed by the second-class passengers, and the third-class passengers had the worst chances of survival. This gradient shows how socioeconomic status affects things. Passengers who were richer were more likely to be closer to lifeboats and get help during the evacuation.

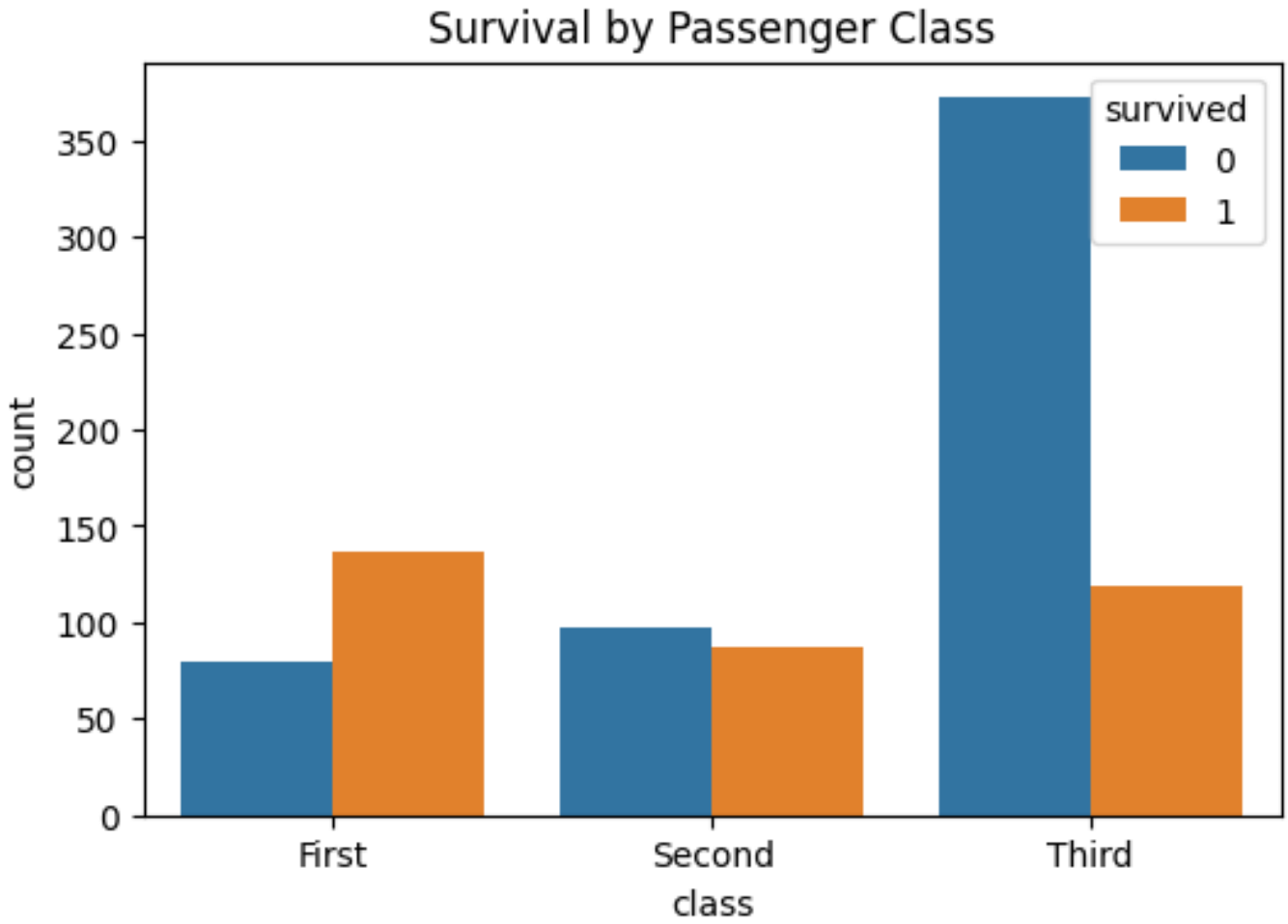

Figure 4: Survival counts by passenger class, showing higher survival rates in first class.

A closer look at survival rates by class backs up this trend. About two-thirds of first-class passengers lived, but less than half of second-class passengers and only about one-fourth of third-class passengers lived. This shows a clear and steady link between class and survival.

Correlation Analysis

Correlation analysis gives a numerical value to how variables are related to each other. The correlation matrix shows that passenger class has a moderate negative correlation with survival. This means that lower classes are linked to lower chances of survival. Fare has a positive correlation with survival, which supports the idea that passengers who paid more had a better chance of surviving.

Age has a weak negative correlation with survival, which means that it isn't a very good predictor on its own. Age may still interact with other variables, such as gender, to affect results, though. The correlation analysis shows that socioeconomic factors are better at predicting survival than demographic factors alone.

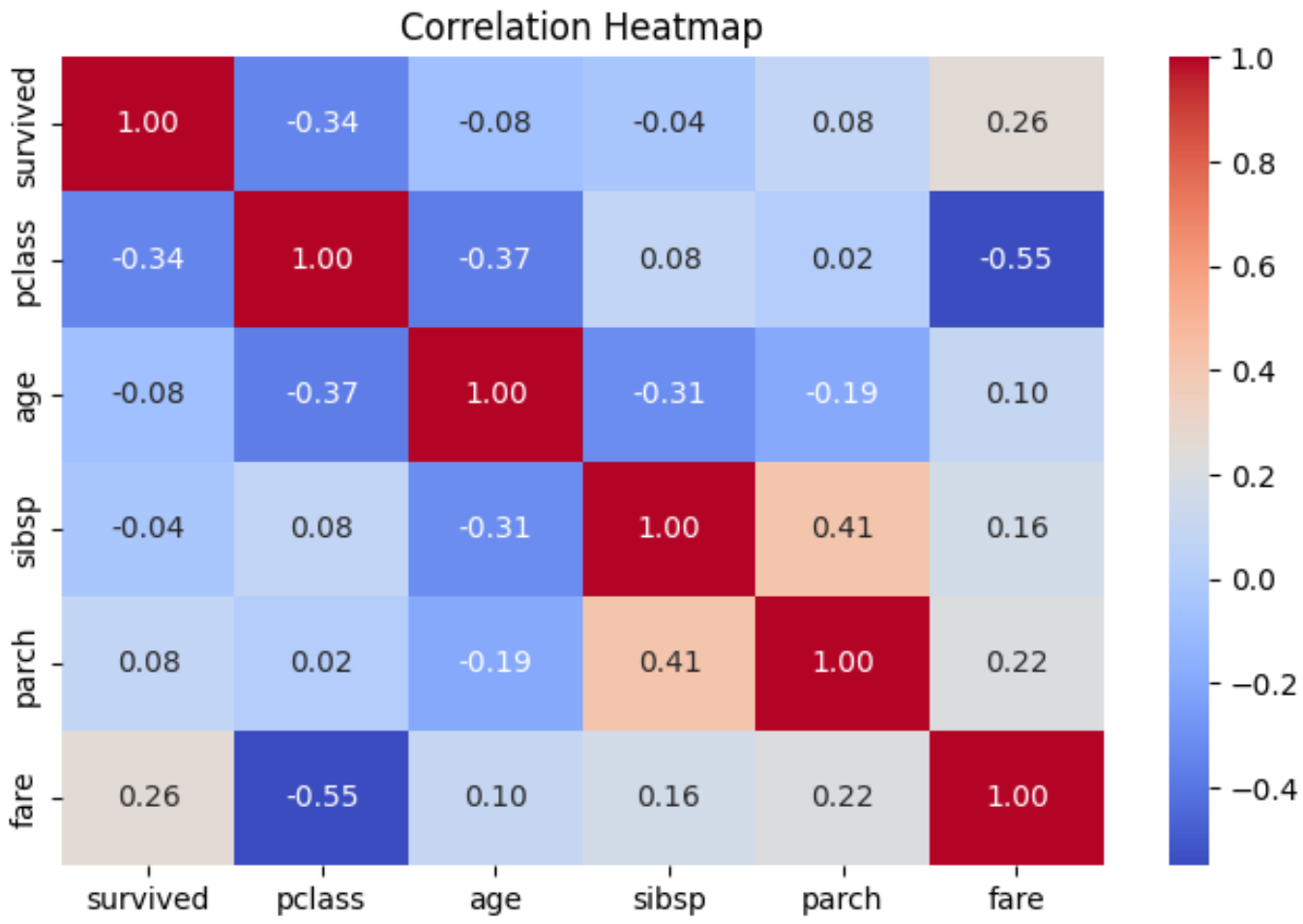

Figure 6: Correlation heatmap illustrating relationships between features and survival.

Machine Learning Models

We used two classification models on the dataset: Decision Trees and Logistic Regression. We chose these models because they show different ways to classify things and give us different kinds of information.

Using a logistic function, logistic regression is a statistical model that figures out how likely it is that a binary outcome will happen. Many people use it because it is easy to understand and use (Hosmer et al., 2013). A coefficient is given to each feature in the model to show how it affects the chance of survival. Positive coefficients raise the chances of survival, while negative coefficients lower them.

Decision Trees, on the other hand, are non-linear models that split the data into groups based on the values of the features. They can show how different variables affect each other in complicated ways, and they are easy to understand visually (Breiman et al., 1984). Decision Trees are flexible for many problems because they don't need you to make any guesses about how the data is spread out.

Model Evaluation

We looked at a number of different metrics to see how well the model worked, such as accuracy, F1-score, confusion matrix, and the area under the ROC curve. These metrics give a full picture of how well the models do at classification tasks.

The confusion matrix shows that the Logistic Regression model did a good job of classifying a lot of passengers. There are a lot of true positives and true negatives, but not many false positives and false negatives. This shows that the model can tell the difference between people who live and people who don't.

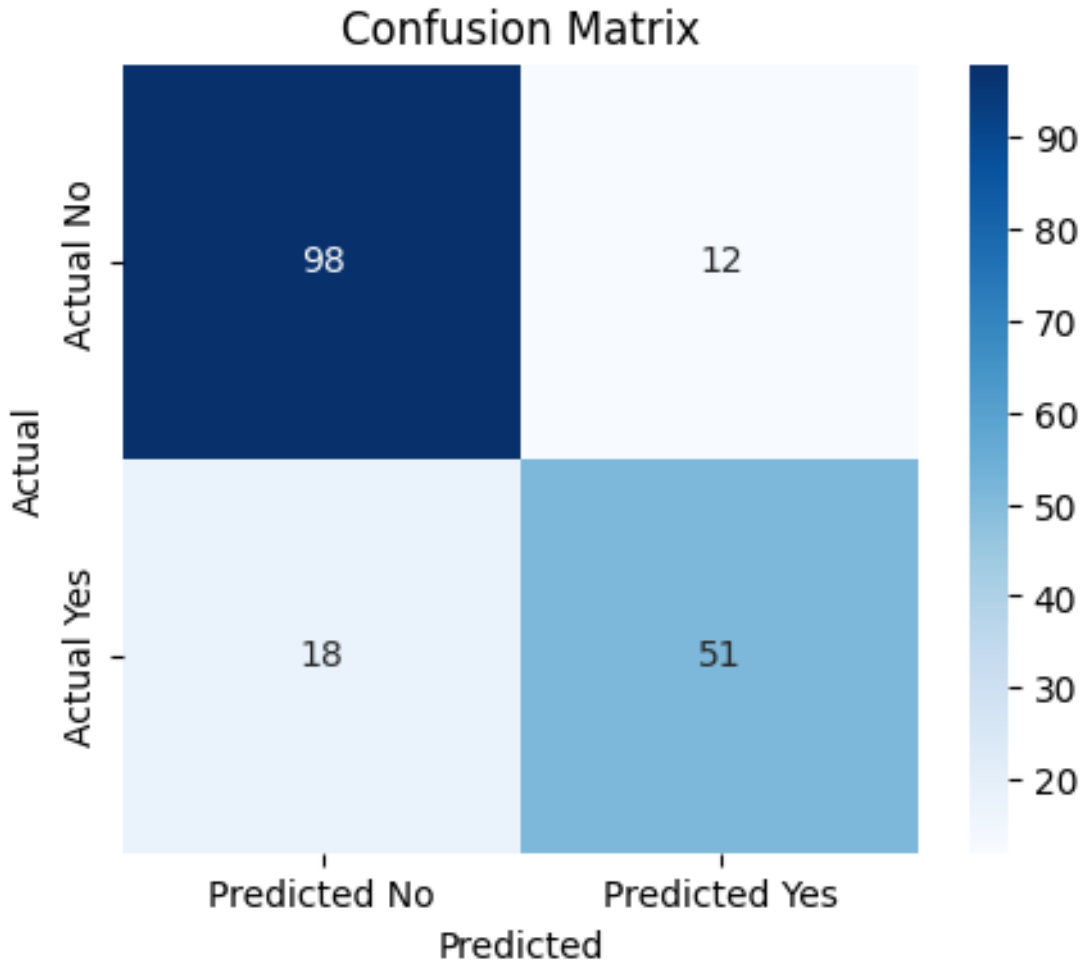

Figure 7: Confusion matrix showing classification performance of the Logistic Regression model.

The ROC curve backs up this conclusion even more. The area under the curve is about 0.87, which shows that it is very good at making predictions. A value close to one means that the model can clearly separate the two classes, while a value close to 0.5 means that the model is not doing a good job (Fawcett, 2006).

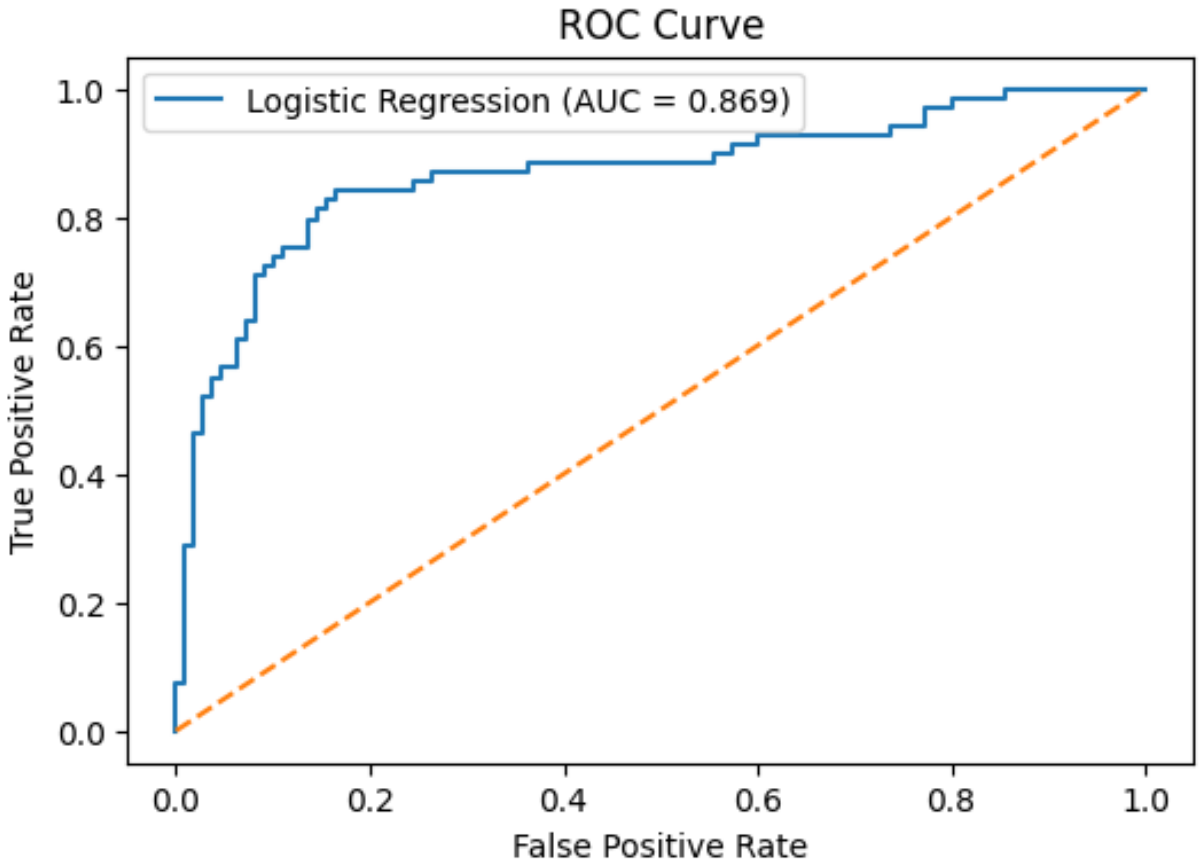

Figure 8: ROC curve for Logistic Regression model with an AUC of approximately 0.87.

Logistic Regression and Decision Tree models both give similar results when it comes to accuracy and F1-score. This means that the dataset isn't very complicated and that both linear and non-linear models can find the patterns that are there. In these situations, simpler models like Logistic Regression may be better because they are easier to understand.

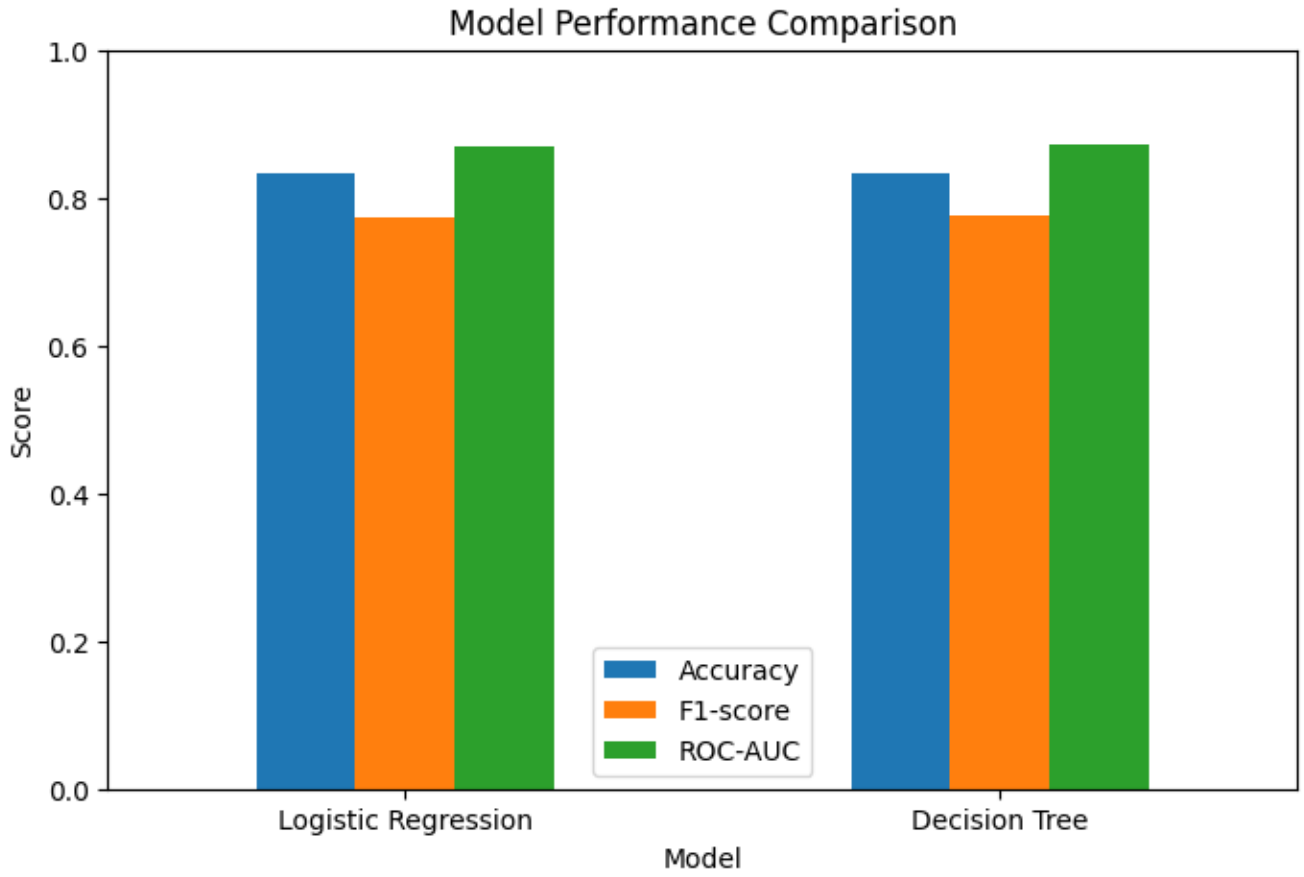

Figure 9: Comparison of model performance metrics for Logistic Regression and Decision Tree.

Feature Importance

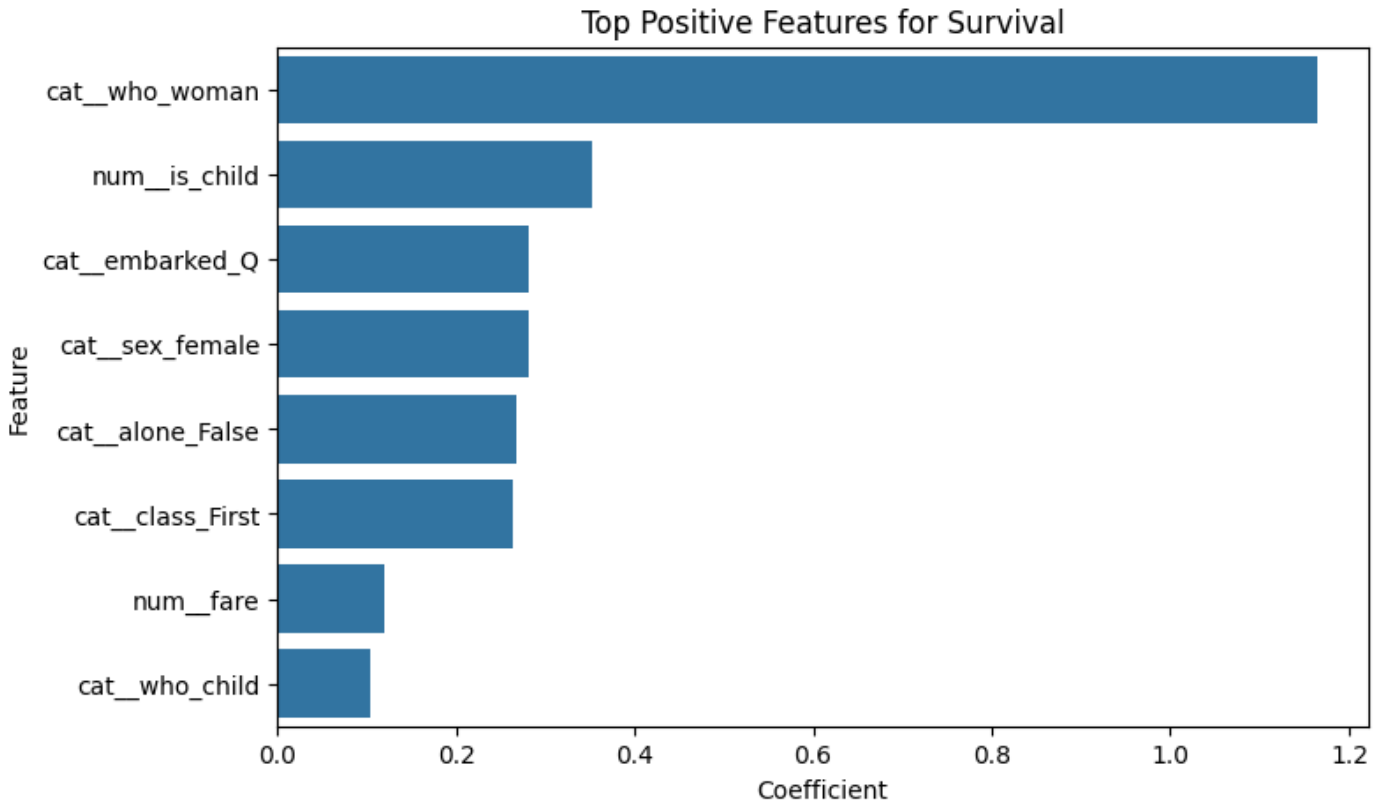

Figure 10: Top positive features contributing to survival prediction.

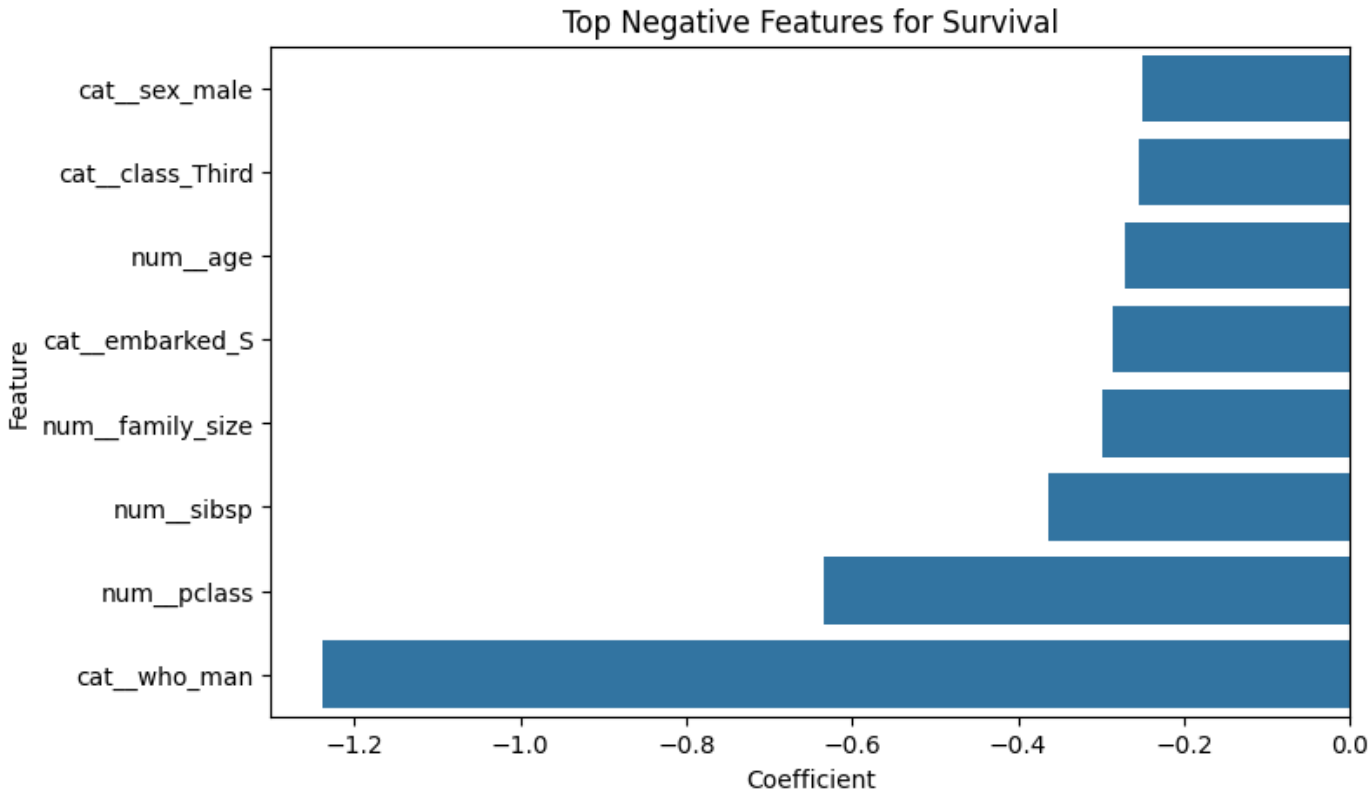

Figure 11: Top negative features reducing the likelihood of survival.

To understand machine learning models, you need to know which features affect predictions. In Logistic Regression, the size and direction of the coefficients tell you how important a feature is.

The study finds that being a woman is the best sign that you will live. Being a child, paying a higher fare, and being in first class are other good things. These factors make it more likely that someone will survive, and they fit with what has been said about the disaster in the past.

Being a man, being in the third class, being older, and having a bigger family are all bad signs. These things make it less likely that someone will survive and show how some groups have it worse than others. The strong negative effect of being male shows that women and children were given priority during the evacuation.

Discussion

This analysis shows that a mix of social, economic, and demographic factors affected who lived and who died on the Titanic. Gender and class are the most important factors that affect this, as they show both social norms and structural inequalities. Richer passengers had better access to resources and safer places, and women and children were given priority during the evacuation.

The results show that simple models can do well when the relationships in the data are clear, which is interesting from a machine learning point of view. Logistic Regression helps you understand how important each feature is, and Decision Trees let you see how different variables interact with each other.

The fact that the two models worked almost the same way shows that the problem doesn't need very complicated algorithms. Instead, careful analysis of the data and choice of features are all that is needed to make accurate predictions.

Limitations

Even though the results are very good, there are some limitations that need to be recognised. The dataset has some missing values, especially in the age variable, which could make the model less accurate. The sample size is rather small, which makes it hard to apply the results to other situations.

Also, the dataset doesn't include all the things that could have affected survival, like when the evacuation happened or how the crew acted. Different choices for feature engineering can also change the results, and different ways of preprocessing data can lead to different conclusions.

Conclusion

This report has shown how machine learning can be used to guess who will live on the Titanic. Several important insights were found through exploratory data analysis, correlation analysis, and model evaluation. Gender and passenger class were the most important factors. Women and first-class passengers had much higher survival rates.

The AUC for both the Logistic Regression and Decision Tree models was about 0.87, which is very good. These results show how well machine learning can find patterns in data and make accurate predictions. The study also stresses how important it is to be able to understand the data and the real-world context behind it.

In conclusion, this study shows that machine learning can be used to not only predict outcomes but also give us useful information about the things that affect those outcomes. By using both statistical analysis and knowledge of the field, you can learn more about complicated events that happen in the real world.

References

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